

Project revamp - measured results, October 7, 2026

Both repositories were changed locally. Code, immutable evidence, portable replay assets, and editable resume wording are included. Existing Humanoid changes were preserved; nothing was committed, pushed or published.

New measurements cover simulation regression engineering, generated file lookup, real neural HTTP service behavior, external retrieval domains, and deterministic approval/citation contracts. The existing 3,600/384 confirmatory results were verified from saved artifacts and documented; they were not rerun during this revamp.

Humanoid: reproducibility and measured execution

H2: portable real-checkpoint bundle, 30 hashed source/asset files. Relocated from /tmp with PYTHONPATH unset. Five of five nominal repeats matched exactly; 40 of 40 injected regressions were caught. These gates detect changes relative to a reference, not unsafe behavior or physical-robot robustness.

Cached versus uncached validators matched 208/208 tested decisions. Two counterbalanced 5,000-call comparisons reduced median validated execution latency 49-69%. A separate 10,000-call measurement gave 0.313 ms median, 1.092 ms p99, 16.277 ms max and zero observed misses against a 40 ms policy period. Intel Xeon Platinum 8480CL host; warm CPU execution excludes physics, I/O and startup. The speedup is for added validation, not the original unguarded controller or faster ONNX inference.

H1: a fresh eight-episode development pilot evaluated one actor on two layouts, nominal sensing and 35% dropout. Both baseline and existing yaw filter reached 2/4 goals without falls. Command flip rates fell about 53%; no goal-success or fall-rate improvement is established. These layouts are now consumed development data.

Saved confirmation: flat falls off/default 77/360 versus 0/360; recorded push falls 341/360 versus 13/360. Navigation confirmation completed 384 episodes and retained its NEGATIVE zero-fall verdict. Results use a 12-DoF biped in simulation and oracle pose/goal for navigation.

MetaNavT: real data, external validation and service behavior

M1: 600 unique actual experiment files, 110 source groups, 360 generated scalar lookup questions (120 dev/240 test). Each labeled split spans 13 disjoint experiment directories. Snapshot, question, manifest and executing-source hashes are recorded. Body-only versus path-context BM25 test Recall@10 is 0.058 versus 0.671; nDCG@10 is 0.024 versus 0.485, paired family-bootstrap delta +0.461 [0.297, 0.666]. These labels are mechanically generated, with zero independent human reviews; they do not establish answer or newest-version accuracy.

M3: full official NFCorpus and FiQA test splits total 61,271 documents and 971 queries. Graded nDCG and positive-relevance recall were independently checked against pytreval. NFCorpus hybrid beats BM25 (0.3637 versus 0.3207), but bounded reranking reduces it to 0.3555. FiQA dense beats BM25 (0.3848 versus 0.2369), while hybrid falls to 0.3645. Fixed configurations show domain-dependent tradeoffs; no automatic per-domain selector has been validated.

Two separate same-GPU ablations: fp32 to fp16 at fixed 512 tokens/top 20 lowers NFCorpus warm offline p95 216.52 to 77.15 ms, with paired nDCG delta -0.0002 [-0.0004, about zero]. Default 8,192 to 512 tokens at fixed fp32/top 20 lowers p95 637.44 to 226.57 ms, with nDCG delta +0.0006 [-0.0025, +0.0040]. Neither interval establishes formal quality equivalence; index/model load and HTTP are excluded.

M2: the original FastAPI/PostgreSQL16/pgvector SQL service indexed 600 real files into 5,633 nodes. H100 GPU fp16 client p95 is 252/422/1,319 ms at concurrency 1/4/16, with 16 requests per level; all 48 measured requests plus cold probe and two warmups succeeded. Generated first-16 lookup questions measure latency, not answer accuracy or capacity. PostgreSQL ran via task-local Apptainer; the Docker GPU override was parsed but not executed.

Matched fp32 CPU/GPU profiles used the same 61-file fixture and 16 queries at each concurrency. GPU p95 is 62/235/889 ms, with zero errors. CPU succeeded at concurrency 1/4, then 13/16 requests timed out at concurrency 16 (120 s timeout); failures remain in all-attempt statistics. Different hosts/devices prevent attributing this comparison to precision or software changes alone. Interrupted exploratory CPU accounting and failed SQL_ASCII startup are retained; UTF8 initialization fixed the latter for the real-file run.

MetaNavT: verifiable file edits and evidence contracts

M4: trusted operator CLI, immutable content/path/session-bound capabilities, short expiry, SQLite single-use consumption/audit, descriptor-based regular-file operations, no implicit overwrite, and frontend review export/token submission. Boolean approval bypasses and an in-process library key-reader route were removed. Header/filename newline injection, wrong-source value borrowing and CRLF corruption were also corrected.

220/220 malicious action fixtures were denied without mutation across 22 families; 100/100 benign controls passed (80 approved writes and 20 reads). A separate 240-case synthetic JSON-fact suite passed 80 supported, 80 conflict and 80 unknown cases. These are deterministic enforcement/structured-fact contracts, not LLM prompt-injection resistance, human citation accuracy or natural-language entailment. Protect operator keys against every other tool; same-account arbitrary OS execution remains outside this boundary. Physical citation mapping across all production reader transformations remains a follow-up validation item.

Validation and practical limits

MetaNavT broad eval/unit/engine/API/MCP run: 299 passed, three optional skips. After review fixes, 89 integrity tests passed with one Octave runtime skip; final M4 focused suite passed 118 with one skip; API/SDK passed seven. Twelve optional TREC tests and 19 load/BEIR/replay tests passed. Existing fixture gate passed all six configurations with unchanged ranking metrics. Frontend production build, lint/types, static export and formatting passed; lockfile unchanged.

Humanoid full CPU run: 2,105 passed, two stale assertion failures, three skips. Narrow assertion corrections preserved frozen clock behavior and exact task-registration checks; both affected tests then passed. Nine actual-asset integration checks and three mocked CI-intake checks passed. The full suite was not rerun after assertion fixes. Fresh dependency installation, GitHub CI execution and second-host determinism are unverified.

Academic basis and employer fit

BEIR and its reproducible-reference work informed full-domain evaluation, graded labels, TREC artifacts and paired deltas. The statistical-precipice paper informed grouped uncertainty and separate per-policy failure reporting. ALCE informed separating source resolution, literal support and entailment. AgentDojo informed adversarial/benign utility controls; the new suite enforces capabilities without claiming to reproduce AgentDojo LLM experiments.

As an inference from the researched postings: Humanoid replay/fault/timing evidence fits Neuralink Simulation and robotics test work; dataset provenance, pgvector operation and load measurements fit Torc MLOps/Data Engineering and Appttronik infrastructure; retrieval and trusted action execution fit SpaceX Applied AI. Rust/C++ and professional/hardware experience requirements remain qualification gaps where applicable; project metrics do not replace them. Keep six-person capstone retrieval/data-API credit separate from these independent 2026 enhancements.

Remaining measured-result tasks

M1: independently review/rephrase questions and add temporal conflict, multi-artifact and unanswerable cases. Validate every ingestion adapter against physical source bytes before claiming citation accuracy across formats. The review CSV is a draft, not completed human labeling.

H1: lock a candidate and fresh-layout protocol before a larger matched navigation confirmation. Do not present the eight-episode chatter result as fall reduction.

M2/M3: repeat load sessions, score retrieval quality through the actual API, and select retrieval configurations on development data before new holdout evaluation. Current domain winners were observed after testing.

H3/H4: history/latency policy training and repeatable 22-DoF turning remain proposed training work. No new PPO training or hardware deployment was performed.

Primary academic sources

- Demonstrating Berkeley Humanoid Lite (RSS 2025)
- Real-World Humanoid Locomotion with Reinforcement Learning (Science Robotics 2024)
- RMA: Rapid Motor Adaptation for Legged Robots (RSS 2021)
- Deep RL at the Edge of the Statistical Precipice (NeurIPS 2021)
- Learning to Walk in Minutes Using Massively Parallel Deep RL (CoRL 2021 / PMLR 2022)
- BEIR: A Heterogenous Benchmark for Zero-shot Evaluation of Information Retrieval Models
- Resources for Brewing BEIR: Reproducible Reference Models and Statistical Analyses
- Fact or Fiction: Verifying Scientific Claims
- Enabling Large Language Models to Generate Text with Citations
- FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation
- Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity
- Is ChatGPT Good at Search? Investigating Large Language Models as Re-Ranking Agents
- Lost in the Middle: How Language Models Use Long Contexts
- AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents